

# Exponential Expressions

## Lesson Plan

### 1. Lesson Information

|        |                                    |              |              |
|--------|------------------------------------|--------------|--------------|
| Course | Grade 7 Accelerated<br>(1205050)   | Unit         | 3            |
| Lesson | Lesson 3 — Exponential Expressions | Date         | _____        |
| Length | 53 minutes (one period)            | Benchmark(s) | MA.7.NSO.1.1 |

### 2. Standards

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

**Must.** Two or three laws in one expression, whole-number exponents and rational bases; generate an equivalent expression and evaluate it.

**Must not.** No negative exponents yet. The laws never cross addition or subtraction, and order of operations stays at six steps or fewer.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR            | What it asks                                                         | The evidence in this lesson                                                                                                                 |
|----------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR.2.1 | Demonstrate understanding by representing problems in multiple ways. | Notes II and board 9 — one expression is carried to the end by two different plans, written side by side, and the two answers are compared. |
| MA.K12.MTR.3.1 | Complete tasks with mathematical fluency.                            | Boards 1 through 4 — four multi-law expressions, one each, written without a reference sheet.                                               |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions.                              | Board 5 — a sixth power over $2^{10}$ lands on 4; students say why an answer that small is reasonable before they trust it.                 |

### 4. Learning Target and Essential Question

**Learning target.** I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

**Essential question.** How can the laws of exponents be used to generate equivalent numeric expressions?

**Building on.** Lessons 3.01–3.02 — the seven laws; MA.7.NSO.2.1 — order of operations with rational numbers.

**Working toward.** MA.8.NSO.1.3 — the same combined work with integer exponents (3.06–3.07); MA.8.AR.1.1 — with variable bases (Thread A).

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 10        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 14        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (14 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                       | DOK   |
|--------------------------------------------------------------------------------------------------------------------------------|-------|
| Winston works inside the parentheses first and Yvette expands first. Why is the order yours here, but not in $2 + 3 \cdot 4$ ? | DOK 3 |
| $3.2^0$ sits inside a sixth power and changes nothing. What would have to change about that factor for it to matter?           | DOK 3 |
| Which plan for $(0.5^3)^2 \div 0.5^2$ uses fewer digits, and why?                                                              | DOK 2 |

## 8. Checks for Understanding

each board, and what to do about it

| # | What the board asks                                                                                                      | Answer                                          | If the board is wrong                                                                                                                                                                                                                                                                                                         |
|---|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A warehouse holds $6^4$ boxes in $6^2$ equal rows. Each box weighs 3 kilograms. How many boxes are in each row?          | <b>36</b>                                       | 1296 — the $6^2$ rows were ignored [the quotient was never taken]; $6^6$ or 46,656 — the exponents were added [confused with product of powers]; anything built from 3 — the unneeded figure was used                                                                                                                         |
| 2 | $(3^0 \cdot 3^2)^3$                                                                                                      | <b>729</b>                                      | 243 — added the exponents $2 + 3$ in the power of a power step [confused with product of powers]; 0 — zero exponent read as 0                                                                                                                                                                                                 |
| 3 | $\frac{(0.5^3)^2}{(0.5)^2}$                                                                                              | <b>0.0625</b>                                   | 0.125 — divided the exponents $6 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]; 0.015625 — $0.5^6$ , the denominator ignored                                                                                                                                                                           |
| 4 | $\left(\frac{2.5^5}{2.5^3}\right)^1$                                                                                     | <b>6.25</b>                                     | 2.5 — the outside 1 was treated as the final exponent [identity law misapplied]; 15.625 — used $2.5^3$                                                                                                                                                                                                                        |
| 5 | $\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$ Which is the value?                                                           | <b>D — 4</b>                                    | (A) $(2^2)^6$ written as $2^8$ by adding $2 + 6$ , then $2^8 \div 2^{10}$ [power of a power confused with product of powers]; (B) added the exponents $12 + 10$ instead of subtracting [confused with product of powers]; (C) $3 \cdot 2^0$ read as 0, so the whole numerator is 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$ ] |
| 6 | $\frac{0.2^3}{2^3} \cdot 0.1^2$ Which expression is equivalent?                                                          | <b>C — 0.1<sup>5</sup></b>                      | (A) quotient of powers applied to DIFFERENT bases, 0.2 and 2 [7.NSO.1.1: the quotient law needs the same base]; (B) the $2^3$ in the denominator was ignored [order of operations: the fraction bar is a grouping]; (D) multiplied the exponents $3 \cdot 2$ [confused with power of a power]                                 |
| 7 | $0.1^2 \cdot (0.1^1)^2$                                                                                                  | <b>0.0001</b>                                   | 0.001 — $0.1^3$ , the outer square on $(0.1^1)^2$ dropped, leaving $0.1^2 \cdot 0.1^1$ [power of a power step skipped]; 0.01 — stopped at $0.1^2$                                                                                                                                                                             |
| 8 | $\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$                                                                                     | <b>-64</b>                                      | 64 — dropped the sign; 'undefined' — $(-4)^0$ in the denominator read as 0, so a division by 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$ ]                                                                                                                                                                                     |
| 9 | Winston writes $(4^7/4^4)^2 = (4^3)^2$ . Yvette writes $(4^7/4^4)^2 = 4^{14}/4^8$ . Do they get the same value? Explain. | <b>Yes — both give <math>4^6 = 4,096</math></b> | 'No' — judged equivalence by appearance instead of value [7.NSO.1.1: generate equivalent expressions]                                                                                                                                                                                                                         |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

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**ESE / IEP.** The seven laws from Notes I stay on the desk as a half-sheet. Before writing, the student says the plan out loud in two words — 'inside first' or 'expand first' — and the teacher confirms the plan, not the answer.

**ELL.** Plan and evaluate are the two verbs on every slide today. Write PLAN = decide the order and EVALUATE = find the number on the board, and ask for the plan in words before any board goes up.

**Enrichment.** Ask for a single power of 2 equal to  $(2^3 \cdot 4^2) \div 8$ , then for one expression that two different laws can both finish, to the same value.

## 10. Closure, Vocabulary, Materials, Practice

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**Closure.** Board 9 is written work and it is the exit evidence: the student has to carry both plans to the end and show they land on the same value. Read the boards, not the papers.

**Vocabulary.** **equivalent expressions; evaluate; plan**

**Materials.** Whiteboards and markers. Calculators are allowed on everything; the decimal values in the bank (5.0625, 2,293.45007, 9,605.9601) are calculator work once the single power is found.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate expressions using properties of exponents — ZE9.